



LearnMore

Daily Tech-Skill Trainer for Windows

User Manual · Version 1.2

One new lesson, every single day.

Data Structures & Algorithms · Design Patterns · SQL Server Optimization ·
System Design · Frontend Engineering

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1. What is LearnMore?

LearnMore is a Windows desktop application built for one purpose: to make sure you learn **one new technical thing every day**. It assigns you a single daily lesson, reminds you with Windows notifications until you finish it, and tracks your streak so skipping a day hurts.

The built-in library contains **~150 lessons** across five tracks:

Track	What you'll learn
■ Data Structures & Algorithms	Big-O, arrays, hash tables, trees, graphs, sorting, dynamic programming — with C# examples
■ Design Patterns	SOLID, all GoF patterns, DI, CQRS, resilience patterns — with ASP.NET Core examples
■ SQL Server Optimization	Indexes, execution plans, statistics, parameter sniffing, locking — with runnable T-SQL
■ System Design	Caching, queues, sharding, CAP, sagas, real case studies (URL shortener, chat, feed)
■ Frontend Engineering	Angular signals, RxJS, change detection, CSS grid/flex, web vitals, security

Each lesson includes a **reading** (5–20 minutes), a **practice task** you can apply to real code, **reference links**, and a **quiz**. The day only counts when you pass the quiz.

2. Installation

Requirements

- Windows 10 or 11 (64-bit)
- **SQL Server LocalDB** — usually already installed with Visual Studio or SQL Server Express. Verify with the command below; it should list *MSSQLLocalDB*.

```
sqllocaldb info
```

If missing, install "LocalDB" from the SQL Server Express installer (Microsoft download, free).

Install steps

- Run **LearnMore Setup 1.0.0.exe**.
- Choose an installation folder (or accept the default).
- Finish — LearnMore starts automatically and registers itself to start with Windows.

■ **Note:** On first run Windows SmartScreen may warn about an unknown publisher — the installer is not code-signed. Click **More info** → **Run anyway**.

3. First Launch

On first start the app:

- Starts its local API in the background (nothing is sent to the internet — everything runs on your machine).
- Creates a database named **LearnMore** on your LocalDB instance and fills it with all lessons. This takes a few seconds.
- Assigns **today's lesson** and opens the Dashboard.

✓ **Tip:** The very first lesson is always an easy one — the difficulty grows as you progress through each track.

4. The Dashboard

The Dashboard is your home screen. It shows:

- **Today's lesson card** — the topic, title, estimated minutes and difficulty (★ = easy, ★★★ = hard). Click **Start learning** → to begin.
- **Current streak** ■ — consecutive days completed. Miss a day and it resets to zero.
- **Longest streak** ■ — your record. Beat it.
- **Last 30 days calendar** — green = completed, red = missed, yellow = today, still pending.

The sidebar is always visible: **Dashboard**, **Today** (with a yellow dot when the lesson is pending, green when done), **Topics**, and **Stats**.

5. Today's Lesson — Your Daily Routine

This is the page you'll use every day. Work through it top to bottom:

Step 1 — Read the lesson

The main card contains the lesson with code examples and tables. Reading time is shown in the header (typically 12–22 minutes).

Step 2 — Do the practice task

Below the lesson is a ■ practice task — a small, concrete exercise, usually pointed at your own real projects. This is where the knowledge actually sticks. The app doesn't check it; it's on your honour.

Step 3 — Pass the quiz

- Answer every question (the Submit button stays disabled until all are answered).
- **All correct** → the day is complete ■ — your streak increases and the tray icon tooltip flips to “done for today”.
- **Wrong answers** → they are highlighted in red, the correct answer in green, and an explanation appears. Read it, then resubmit — you can retry as many times as you like.

Step 4 — Go deeper (optional)

The  links open the official docs or the best article on the subject in your browser.

✓ **Tip:** Completed lessons stay readable forever — revisit them any time from the Topics page, with quiz answers revealed.

6. Arabic Explanations (الشرح بالمصري)

Every lesson includes a condensed explanation written in **Egyptian colloquial Arabic** — the way developers actually explain things to each other. Technical terms (index, cache, closure, deadlock...) stay in English; the explanation around them is مصري .

Using it

- On any lesson page (Today, or a completed lesson), click the  اشرح بالمصري button in the lesson header.
- The Arabic panel appears below the English lesson, right-to-left, with code snippets kept left-to-right.
- Click again (إخفاء الشرح) to hide it. Your choice is remembered — if you leave it on, every lesson opens with the Arabic explanation visible.

How to learn with it

Recommended flow: read the English lesson first (the terminology you'll use in interviews and documentation is English), then read the Arabic explanation to lock in the intuition. The quiz stays in English on purpose — that's the vocabulary being trained.

✓ **Tip:** Adding your own lessons? Put Arabic explanations in `seed\ar\*.json` files matched by lesson title — see the README for the exact format.

7. Assessments, Roadmap & Courses

Assessment — find your level

Each track has an interview-style assessment (~20 multiple-choice questions across Junior/Mid/Senior difficulty). Your level per track:

Level	Requirement
Junior	Junior-tier questions $\geq 70\%$
Mid-Level	Junior achieved AND mid-tier $\geq 60\%$
Senior	Mid achieved AND senior-tier $\geq 60\%$

After submitting you get: your level, a per-tier score breakdown, every missed question with its explanation and a link to the lesson that teaches it, and **course recommendations** for reaching the next level (free, paid  and Arabic  options). Retake assessments monthly to measure progress.

Roadmap — see the whole climb

- Every track shown as a Junior → Mid → Senior ladder of lessons.
- ■ completed lessons, ■ today's lesson, ■ upcoming.
- ■ YOU ARE HERE marker placed from your latest assessment result.
- ■ weak-spot flags on lessons matching questions you missed — the app literally shows you what to study next.

✓ **Tip:** Flow that works: take the assessment → check the roadmap → let the daily lessons close the gaps → retake in a month and watch the marker climb.

8. Browsing Topics

The Topics page shows the five tracks with progress bars. Click one to see its full lesson list in learning order:

Icon	Meaning
■	Completed — click to re-read it, including quiz answers and explanations
■	Today's lesson — click to jump to the Today page
■	Locked — will unlock when the daily engine assigns it on a future day

Locking is deliberate: the app enforces **one lesson per day**. You cannot binge the whole track in a weekend and forget it by Monday — spaced daily learning is the entire point of LearnMore.

9. Stats & Streaks

- **Current / longest streak** and **total lessons completed**.
- **Percentage of the whole bank** — ~150 lessons ≈ 5 months of daily learning.
- **Per-topic progress bars** — the engine rotates topics evenly, so these grow together.

Streak rules

- Complete today's quiz → streak +1 (if yesterday was also completed).
- Miss a day → streak resets to 1 on your next completed day.
- Longest streak is never lost — it's your all-time record.

10. Reminders, Tray & Auto-Start

New in v1.1 — motivation, not just nagging: streak-milestone celebrations (3, 7, 14, 30, 60, 100+ days ■), a congratulation toast when you finish the day, near-finish nudges when a track has ≤3 lessons left, and rotating reminder messages.

LearnMore is designed to be impossible to ignore:

- **Starts with Windows**, minimized to the system tray.

- From **09:00** (default), if today's lesson is not done, a Windows toast notification appears — and repeats every **2 hours** until you finish.
- Clicking the notification opens the app directly on today's lesson.
- **Closing the window does not quit** — the app hides to the tray so reminders keep working. To really quit: right-click the tray icon → **Quit**.

Tray icon menu

Item	Action
Open LearnMore	Shows the main window
Today's lesson	Opens the window directly on the Today page
Quit	Fully exits the app (reminders stop until next login)

Hover the tray icon to see the status: “today's lesson is waiting ■” or “done for today ■”.

11. How the Daily Engine Picks Lessons

- **Topic rotation:** round-robin — the topic that was assigned longest ago goes next, so all five tracks advance evenly (roughly each topic every 5 days).
- **Difficulty progression:** within a topic, all ★ lessons come before ★★, then ★★★ — foundations first.
- **One per day:** a new lesson is created the first time the app checks after midnight. Yesterday's unfinished lesson counts as missed.
- **After the bank is finished** (~5 months), lessons recycle oldest-first as review days.

12. Adding Your Own Lessons

Lessons live as JSON files in the **seed** folder inside the installation directory (*resources\seed*). Each file is one topic:

```
{
  "topic": "SQL Server Optimization",
  "color": "#ef4444",
  "icon": "■",
  "items": [
    {
      "title": "My New Lesson",
      "difficulty": 2,
      "estimatedMinutes": 15,
      "bodyMarkdown": "# Heading\n\nLesson text with **markdown** and ```sql\ncode```...",
      "practiceTask": "Try it on your own database.",
      "externalLinks": ["https://..."],
      "quiz": [ { "question": "...?", "options": ["A","B","C","D"],
        "correctIndex": 0, "explanation": "..." } ]
    }
  ]
}
```

```
]
}
```

- Add new items to an existing file (existing topic) or create a new .json file (new topic).
- Restart the app — the seeder adds anything with a **new title**; existing lessons are never modified or duplicated.

13. Automatic Updates

LearnMore updates itself. On start (and every 4 hours) it checks github.com/mohamedanter1996/learnMore releases, downloads new versions in the background, and installs when you quit the app.

- A toast appears when an update is downloaded — keep working; it installs on quit.
- Or right-click the tray icon → ■■ **Restart to update** to apply immediately.
- Your progress is always preserved: database migrations and new lessons apply automatically on the updated version's first launch.

14. Study Plans, Mind Map & Tech News

■ Study Plans (with calendar tracker)

Beyond the guided daily lessons, you can set your own learning goals over a time period and track them — perfect for prepping a course, a certification, or a new skill.

- Open **Study Plan** in the sidebar → **New plan**: give it a title and a start/end date.
- Inside the plan, add **goals** (free text — anything you want to learn) and check them off as you achieve them.
- Use the **calendar** to click each day you actually studied — the plan tracks your goal %, days studied, study-day streak, and days remaining.

✓ **Tip:** Combine it with the daily lessons: let the app pick your daily lesson, and use a plan to drive a bigger goal like 'Finish the SQL optimization track by month-end'.

■ Roadmap mind map

On the **Roadmap** page, use the **List / Mind map** toggle (top right). The mind map draws each track as a tree — the topic at the center, Junior/Mid/Senior tiers branching out, and every lesson as a dot: green = completed, blue = today, grey = upcoming, with a red ring on your assessment weak spots ■. Click a dot to open that lesson. Your preferred view is remembered.

■ What's New in Tech

The **What's New** page keeps you informed of developments in the tools you use — .NET, C#, Angular, TypeScript, SQL Server, EF Core, and the web platform. Each technology lists recent highlights and a 'learn next' pointer. The feed is refreshed with every app update.

■ Notification settings

The **Settings** page (■) lets you set your reminder time, how often reminders repeat (every 1–4 hours), and toggle notifications. You'll also get a morning kickoff and an evening streak-saver, with rotating bilingual (■ / English) encouragement so it never feels repetitive.

15. Troubleshooting

Problem	Fix
"API failed to start. Check that SQL Server LocalDB is installed" on launch	Run sqllocaldb info in a terminal. If nothing is listed, install LocalDB (SQL Server Express installer). Then restart the app. You can also try sqllocaldb start MSSQLLocalDB .
Window is blank / stuck loading	The API may still be starting (first run seeds the database). Wait ~15 seconds; if still blank, quit via tray and start again.
No reminder notifications	Check Windows Settings → System → Notifications: notifications must be enabled for LearnMore, and Focus Assist / Do Not Disturb off. Also confirm the app is running in the tray.
App doesn't start with Windows	Task Manager → Startup apps → enable LearnMore .
Start over with a fresh database	Quit the app, run: <code>sqlcmd -S "(localdb)\MSSQLLocalDB" -Q "DROP DATABASE LearnMore"</code> then start the app — it recreates and reseeds everything (progress and streaks are lost).
Where is my data?	Everything is local: a SQL Server LocalDB database named LearnMore under your Windows user. Nothing is uploaded anywhere.

16. FAQ

Can I do more than one lesson per day?

No — by design. One focused lesson daily beats binge-reading. You can, however, re-read any completed lesson from Topics.

What happens if I miss a day?

The missed day shows red in the calendar and your streak resets. The lesson you missed isn't lost — it stays in the pool and will be assigned again on a future day.

Do wrong quiz answers hurt my streak?

No. Wrong answers show explanations and you can retry immediately — the goal is learning, not punishment. Only leaving the day unfinished hurts.

Can I change the reminder time?

The default is 09:00, repeating every 2 hours. These are stored in the database (Settings table) and exposed via the local API (PUT <http://localhost:5199/api/settings>). A settings screen is a planned

improvement.

Does it need internet?

No. All lessons are stored locally. Internet is only needed if you click the external “Go deeper” links.

Happy learning — keep the streak alive! ■